

Chapter 6

Contemporary Issues in Engineering

Globalization and Cross Culture Issues

- Process of extending relationship across world. (Such as in terms Economics, Culture, Technology, etc)
- **Globalization** deals with (as per International Monetary Fund)
 - Trade and Transactions
 - Capital and Investment
 - Migration and movement
 - Dissemination of Knowledge

Public Private Partnership (PPP)

- Contractual agreement between Public Entity and Private Partners for delivery of infrastructure or service of public interest.

PPP For

Risk management or minimization

Value for money maximization

Equal Participation **Public and Private**

Public – Output (Outcome and Impact) and

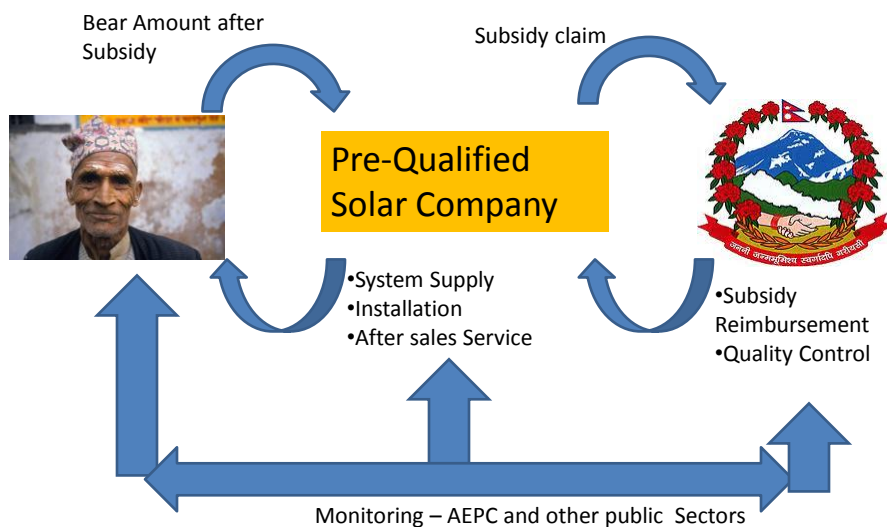
Private – Input (working partners and implementer)

Importance of PPP- means of mainstreaming Economic activities

PPP Model in Nepal

3

PPP model for RE Promotion in Nepal



Safety Risk and Benefit Analysis

Safety is the state of Being Safe in terms of Physical, social, Spiritual, financial, political, emotional, psychological, etc.

Safety first or safe way is the best way

What is Risk?

The probability that a particular threat will exploit a particular vulnerability

- ❑ Need to systematically understand risks to a system and decide how to control them.

5

Risk

The probability of suffering a harm or loss. It is a combination of hazard and Probability

- Risk = Probability of occurrence of hazard X magnitude of hazard

❑ Type of Risks

- Background risk
- Incremental risk
- Total risk

❑ Measurement of Risk

- Individual Risk
- Societal Risk

Individual Risk

□ Risk posed by an individual

- It is generally measured in time domain for a particular type of hazard.
- Risk of Death due to fire in an industry = $1.0\text{E-}05$ /yr

Societal Risk

□ Risk for a group of people

- It is measured in terms of number of people poses particular risk for a given period of time.
- Risk of Death to 10 workers due to fire in an industry = $1.0\text{E-}09$ /yr/10 persons

Acceptable Risk

- ❑ Risk that's acceptable to regulatory agency and also to the public
- ❑ **Level of Risk of Death Per Year (voluntarily acceptable by public)**
 - Smoking 30 cigarettes per day 1 in 200
 - Man aged 35-44 1 in 600
 - Motor Vehicle Accident 1 in 10 000
 - Accident at home 1 in 12 000
 - Rail accident 1 in 420 000

What is Risk Analysis?

- ❑ The process of identifying, assessing, and reducing risks to an acceptable level
 - Defines and controls threats and vulnerabilities
 - Implements risk reduction measures
- ❑ An analytic discipline with three parts:
 - Risk assessment: determine what the risks are
 - Risk management: evaluating alternatives for mitigating the risk
 - Risk communication: presenting this material in an understandable way to decision makers and/or the public

Benefits of Risk Analysis

- Assurance that greatest risks have been identified and addressed
- Increased understanding of risks
- Mechanism for reaching consensus
- Support for needed controls
- Means for communicating results

Basic Risk Analysis Structure

❑ Evaluate

- Value of computing and information assets
- Vulnerabilities of the system
- Threats from inside and outside
- Risk priorities

❑ Examine

- Availability of security countermeasures
- Effectiveness of countermeasures
- Costs (installation, operation, etc.) of countermeasures

❑ Implement and Monitor

Quantitative Risk Analysis

- Generally used in Information Security
 - Hard to make meaningful valuations and meaningful probabilities
 - Relative ordering is faster and more important
- Many approaches to performing qualitative risk analysis
- Same basic steps as quantitative analysis
 - Still identifying asserts, threats, vulnerabilities, and controls
 - Just evaluating importance differently

Example 10 Step QRA

- Step 1: Identify Scope
 - Bound the problem
- Step 2: Assemble team
 - Include subject matter experts, management in charge of implementing, users
- Step 3: Identify Threats
 - Pick from lists of known threats
 - Brainstorm new threats
 - Mixing threats and vulnerabilities here...

Step 4: Threat prioritization

- Prioritize threats for each assert
 - Likelihood of occurrence
- Define a fixed threat rating
 - E.g., Low(1) ... High(5)
- Associate a rating with each threat
- Approximation to the risk probability in quantitative approach

15

Step 5: Loss Impact

- With each threat determine loss impact
- Define a fixed ranking
 - E.g., Low(1) ... High(5)
- Used to prioritize damage to asset from threat

16

Step 6: Total impact

- Sum of threat priority and impact priority

Threat	Threat Priority	Impact Priority	Risk Factor
Fire	3	5	8
Water	2	5	7
Theft	2	3	5

17

Step 7: Identify Controls/Safeguards

- Potentially come into the analysis with an initial set of possible controls
- Associate controls with each threat
- Starting with high priority risks
 - Do cost-benefits and coverage analysis (Step 8)
 - Rank controls (Step 9)

18

Step 10: Communicate Results

- Most risk analysis projects result in a written report
 - Generally not read
 - Make a good executive summary
 - Beneficial to track decisions.
- Real communication done in meetings and presentations

19

Insurance

- Equitable transfer of Risk of a loss from one entity to another in exchange.
- Insurance is a financial intermediary for possible losses
- Assurance of effective and quality output with possibility of compensation in alternative form.

20

SWOT Analysis

- SWOT is actually an acronym for Strengths, Weaknesses, Opportunities, and Threats.
- SWOT could be defined as follows:
 - Strengths: What does the organisation do well?
 - Weaknesses: What is wrong now?
 - Opportunities: What possibilities exist?
 - Threats: What can go wrong?

The purpose of SWOT Analysis

- It is an easy-to-use tool for developing an overview of a situation
 - It forms a basis for matching your strategy to its situation
- Strengths and weaknesses are generally **internal** factors that are within your control.
- Opportunities and Threats are generally **external** issues that you can't control.

Conducting a SWOT analysis

There are eight steps required to complete a SWOT analysis and create a SWOT matrix (also known as a TOWS matrix).

1. List external opportunities
2. List external threats.
3. List internal strengths.
4. List internal weaknesses.
5. Match internal strengths with external opportunities and list the resulting Strengths-Opportunities strategies in the matrix chart.
6. Match internal weaknesses with external opportunities and list the resulting Weaknesses-Opportunities strategies.
7. Match internal strengths with external threats and list the resulting Strengths-Threats Strategies.
8. Match internal weaknesses with external threats and record the resulting Weaknesses-Threats Strategies.

At this point in the process, you'll have a significant list of potential strategies. You'll need to weigh the impact of the various factors in your analysis and select the most feasible strategy to implement. Ideally, you'll select a SO strategy, but often you'll need to implement one of the other three types of strategies to overcome a weakness or address a threat before being in a position to implement a S-O strategy.

The SWOT Analysis – a Check list of What to Look for

Potential Internal Strengths	Potential Internal Weakness
• A distinctive competence? • Adequate financial resources? • Good competitive skills? • Well thought of by buyers? • An acknowledged market leader? • Well-conceived functional area strategies? • Competitive advantages • Product innovation abilities? • Proven management? • Insulated from strong competitive pressures? • Others?	• No clear strategic direction? • A deteriorating competitive position? • Obsolete facilities? • Lack of managerial depth and talent? • Poor track record in implementing strategy? • Falling behind in R&D? • Too narrow market image? • Higher unit costs relative to key competitors? • Others?

The SWOT Analysis – a Check list of What to Look for

Potential External Opportunities	Potential External Threats
• Changes in technology on both a broad and narrow scale • Changes in local or state policies • Changes in social patterns, population profiles, lifestyle changes, etc. • Local events	• What obstacles do you face? • What is your competition doing? • Are the required specifications for your job, products or services changing? • Is changing technology threatening your position?

Strengths of Nepal

- **Favorable climate**
- **Manageable size of population and country**
- **Ability of people to acquire or learn new skills and techniques**
- **Increased awareness for enterprise promotion or development**
- **Available local expertise or human resource capacity for economic Management**
- **Hard-working people**

Weaknesses

- Excessive political influence in economic decision-making
- Weak political will to take critical economic decisions
- Weak policy formulation and implementation capacity
- Poor exploitation and utilisation of natural resources
- Inadequate capacity to protect and surveillance natural resources
- High donor dependence
- Pull-Him/Her-Down (PHD) Syndrome
- Poor security network
- Weak industrial and technological base and low capacity for added value
- Poor maintenance culture
- Ineffective enforcement of law
- Over-centralisation of institutions and Weak local government institutions
- High level of corruption and mismanagement
- Poor public sector incentives or remuneration
- Weak competitiveness
- Low domestic savings and investment

Opportunities

- **New Nepal Slogan**
- **Foreign direct investment**
- **Regional economic integration**
- **Donor support and goodwill**
- **Globalization (favorable policies and programs)**
- **External training or capacity building opportunities in economic management**

Threats

- Erosion or loss of private sector confidence in the economy, particularly foreign investors
- Donor fatigue/erosion of donor confidence
- National, regional and global insecurity or instability
- Globalization (unfavorable policies and programs)
- Smuggling, capital flight and money laundering
- Unhealthy or unfavorable global trade and industrial competition
- Brain drain

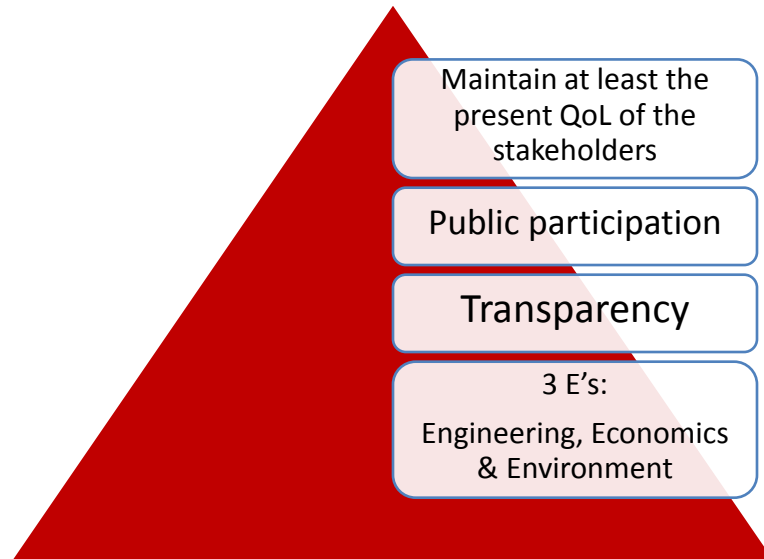
Development and Environment

- Process of Positive Change
- Sustainability and Growth

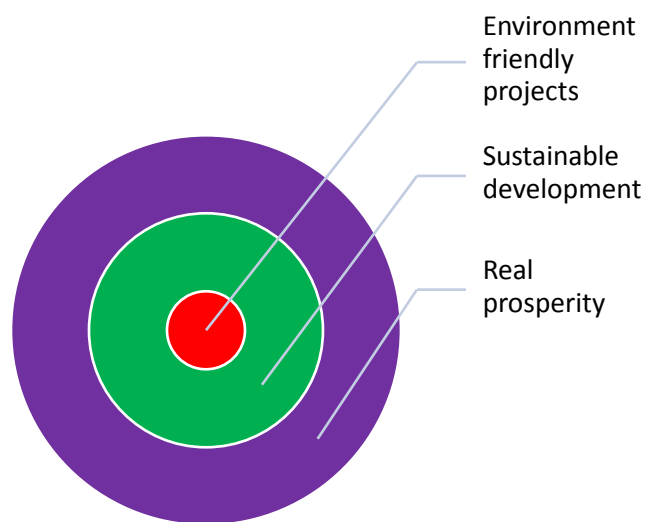
Initial Environment Examination (IEE)

Environment Impact Assessment (EIA)

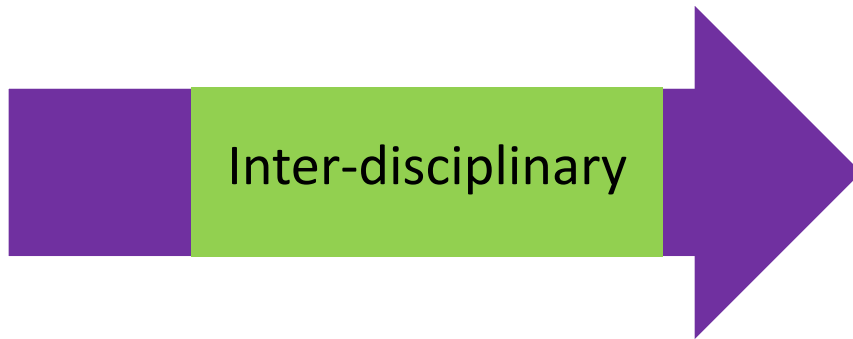
Principle



Aim

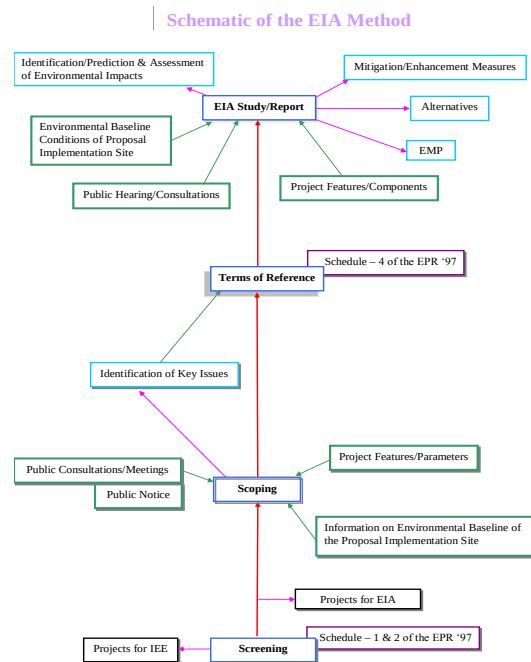


Approach



Key elements

- ☐ Identification of genuine & valid issues
- ☐ Description of the proposal
- ☐ Baseline information on existing environment of proposal implementation site
- ☐ Identification/prediction of impacts and evaluation of their significance
- ☐ Adverse impact mitigation measures/Benefit enhancement measures
- ☐ Environment management plan
- ☐ Analysis of alternatives
- ☐ Feedbacks from public concerned/stakeholders



35 35

Baseline information (cont'd.)

► Socio-economic environment

- ❖ Demography
- ❖ Economic activities/General indication of level of affluence
- ❖ Key local products
- ❖ Access to market & credit facilities
- ❖ Literacy level
- ❖ General price level of key commodities
- ❖ Women/children
- ❖ Vulnerable segment of local populace

36

Baseline information (cont'd..)

❑ Biological environment

- ❖ Forest resources
- ❖ Rare/endangered species of plants
- ❖ Plant species of religious/medicinal/commercial values
- ❖ Wild animals & their habitat
- ❖ Rare/endangered species of animals
- ❖ Community forests
- ❖ Local dependence on forest resources (Timber/Non-timber) for livelihood

37

Baseline information (cont'd.)

❑ Cultural environment

- Temples/Gumbas/Mosques
- Places of historic/cultural significance
 - ❖ Local significance
 - ❖ Regional significance
 - ❖ National significance
- Cemetery/Ghats
- Historic/cultural monuments

38

Mitigation Measures

- Impact-wise MM's with indicative costs
- Avoidance, Corrective or Compensatory

39

Alternative Analysis

- ❖ Design
- ❖ Project site
- ❖ Technology, procedure of operation, time-schedule and raw materials to be used
- ❖ Environment management system
- ❖ Whether or not the risks resulting from the proposal can be accepted
- ❖ Other matters

40

Environmental Management Plan

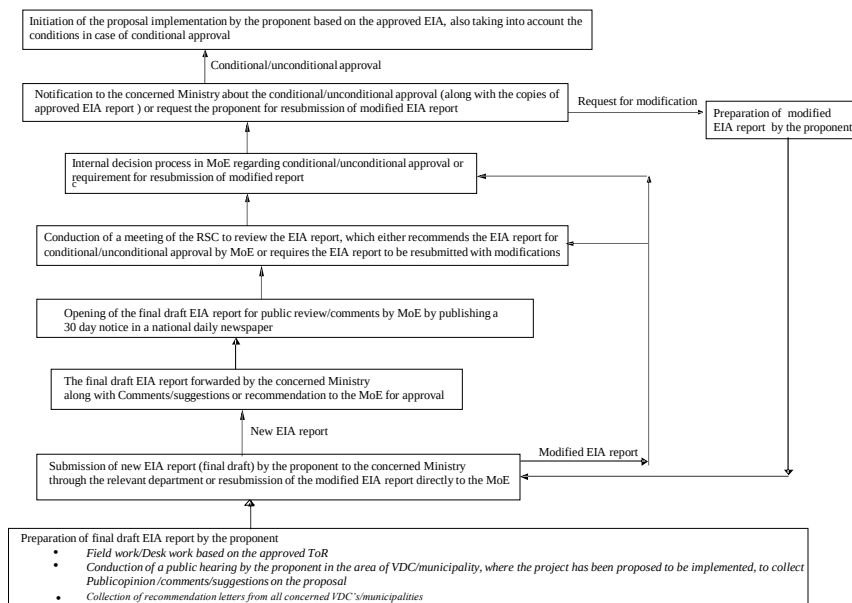
“A plan to ensure systematic integration of all proposed mitigation measures with the project construction/operation activities”

41

EMP: Objectives

- ☐ Formulate project management activities in particular the implementation of EPM's
- ☐ Formulate a monitoring programme for baseline, impact and compliance monitoring
- ☐ Formulate an environmental auditing programme to be implemented after project construction

EIA Report Approval Procedure



Comparison between IEE & EIA

	IEE	EIA
<i>Screening criteria</i>	Schedule-1 of EPR, 1997	Schedule-2 of EPR, 1997
<i>Scoping</i>	Not required	Required
<i>Terms of Reference</i>	Required	Required
<i>Public hearing</i>	Not required	Required
<i>Recommendations of concerned VDC's/Municipalities</i>	Required	Required
<i>Open EIA/IEE report for public review/comments</i>	Not required (However, publication of a 15 day public notice in a national-level daily newspaper and posting of the same in the project implementation area for soliciting suggestions/concerns of all stakeholders is a must before report finalization)	Required
<i>Review by committee of experts</i>	Not required	Required
<i>Approving agency</i>	Concerned agency	MoE
<i>Deadline for approval</i>	Within 21 days of receipt of the report in the concerned agency	Within 60 to 90 days of receipt of the report in the Ministry
<i>EMP</i>	Not required	Required

Conflict and Dispute Management

- Dispute is a one sided declaration of right, claim or demand.
it may be in the form of
 - Cost in time, money and opportunity**
 - Image and behaviors**
 - Employment**
 - Cash flow, etc.**
- Conflict is harmful activities between two person, group, organization, etc.
 - Resolution of Conflict
 - Avoidance (ignoring)**
 - Containment (Bargaining)**
 - Confrontation (face to face solution)**

45

Thanking You

If any Queries Please contact me at
 Email: kayasthasudan@gmail.com
 Mobile No. 9841769047

46